AI/ML Task 2 — Feature Engineering, Model Optimization & Performance Comparison

MainCrafts Technology — AI & ML Internship | Dataset: California Housing (sklearn.datasets.fetch_california_housing) | Target: HousePrice (median house value, $100k units)

Introduction

The California Housing dataset contains 20,640 census-block observations of median housing characteristics. Each row provides eight numeric predictors — MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, Longitude — together with the block's median house value, renamed here to HousePrice per the assignment. The task is supervised regression: predict HousePrice from the eight features and compare several algorithms on the same held-out data.

Methodology

Data acquisition first attempts to load a suitable California Housing CSV from the Task 1 workspace via a schema-validated scan; none is present, so the notebook falls back to fetch_california_housing(as_frame=True) and renames the target to HousePrice (no missing values). Features are separated from the target and split with train_test_split(..., test_size=0.2, random_state=42), yielding 16,512 training and 4,128 test rows. A StandardScaler is then fit on X_train only and used to transform both splits — fitting on the full X before splitting would leak the test distribution's mean and standard deviation into training and inflate the reported metrics. Three regressors are trained on the scaled training data; RMSE and R^2 are computed on the scaled test data. Every number in this document originates from a single executed run.

Feature scaling and Ridge — brief notes

StandardScaler centres and rescales each feature to zero mean and unit variance:

$$z = (x - \mu) / \sigma, \quad \text{with } \mu \text{ and } \sigma \text{ estimated from the training set.}$$

This matters most for the linear models: LinearRegression and Ridge are affected by feature magnitudes, and Ridge's L2 penalty is fair only when features share a comparable scale. Decision trees split on feature thresholds and are essentially invariant to monotonic rescalings, so scaling neither helps nor harms them; a shared preprocessing path is retained for methodological consistency.

Ridge minimises the least-squares loss augmented with an L2 penalty on the coefficients:

$$\text{minimise} \quad \| y - X\beta \|^2 + \alpha \cdot \| \beta \|^2 \qquad (\alpha = 1.0)$$

The penalty shrinks coefficients toward zero and typically reduces variance at the cost of a small bias increase. It is a variance-reduction tool, not a universal overfitting cure — its benefit is largest when features are correlated or when n is small relative to the number of predictors.

Models compared

Linear Regression — ordinary least squares; interpretable linear baseline. Ridge Regression — the same linear model with L2 regularisation (alpha = 1.0). Decision Tree Regressor — a nonlinear, threshold-based partitioner constrained to max_depth = 5, with random_state = 42 for reproducibility.

Results

Model	Test RMSE	Test R^2	Rank
Linear Regression	0.7456	0.5758	3
Ridge Regression	0.7456	0.5758	2
Decision Tree [best]	**0.7242**	**0.5997**	**1**

RMSE is in the same units as HousePrice ($100k), so 0.72 corresponds to a typical error of about $72,000. R^2 is the share of variance in HousePrice explained on the test set — larger is better. The two linear models are practically indistinguishable: at full precision they differ only in the fifth decimal (0.745581 vs 0.745557). This is expected because the eight features are only weakly collinear after standardisation, so L2 shrinkage has little room to help. The Decision Tree is the strongest of the three by a modest but consistent margin — about a 2.86% RMSE reduction and a 2.39-percentage-point R^2 gain over Linear Regression — coming from a handful of nonlinear splits on MedInc, Latitude, and Longitude that the linear models cannot represent. Model selection is performed programmatically by sorting the results by RMSE ascending, then R^2 descending; no model name is hard-coded.

Visualisation

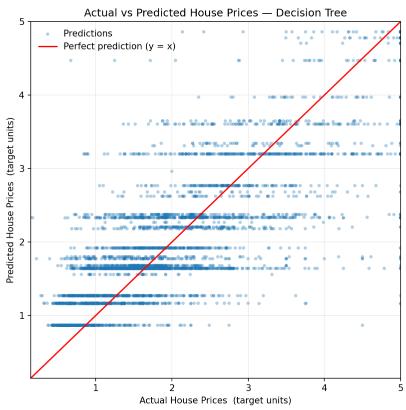

Figure 1. Actual vs Predicted (test set) — Decision Tree. Red line: y = x.

A depth-5 tree produces at most 32 distinct predicted values (2^5 leaves), so the scatter shows clear horizontal banding — predictions collapse onto the 32 leaf means while actual values vary continuously along each band. The main systematic deviation is at the upper end: the target is capped at 5.0 ($500,000), and the tree under-predicts for that ceiling group, so points at the top-right sit below the red reference line. Near the median of the target range, points cluster closer to the diagonal. The residual plot (outputs/figures/residuals_decision_tree.png) has mean −0.0019 and standard deviation 0.724, with a slight fan-out at higher predictions indicating heteroscedastic errors that a deeper model or an ensemble would likely reduce.

Conclusion

Among the three evaluated models under the specified experimental setup — an 80/20 hold-out with random_state = 42, StandardScaler fit on training data only, and hyperparameters fixed to the assignment values — the Decision Tree Regressor (max_depth = 5, random_state = 42) was selected as the best-performing model, with a test RMSE of 0.7242 and a test R^2 of 0.5997. It was chosen by the programmatic selection rule (minimum RMSE; R^2 as tiebreaker) applied to the executed results table. The comparison shows that additional linear regularisation contributes almost nothing over ordinary least squares on this dataset, while even a shallow nonlinear model captures enough of the geographic and income structure to move both metrics meaningfully. The result is not a universal claim that decision trees dominate linear models — only that, at these hyperparameters and on this split, the tree provides the best fit of the three.

Limitations and next steps

Evaluation relies on a single 80/20 hold-out rather than cross-validation, and the three algorithms are all evaluated at fixed hyperparameters with no search. Only RMSE and R^2 are reported — a full study would add MAE and calibration checks. Natural extensions: relax max_depth, add an ensemble (Random Forest or Gradient Boosting), or engineer domain features such as distance to major coastal cities.